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Integration of AI GPTs in music education and their impact on students' perception and creativity

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Abstract

The study aims to investigate the impact of ChatGPT on music education by evaluating creativity and perception indicators resulting from AI integration. The research is based on analyzing the effect of AI (ChatGPT) integrated into music education. To this end, a method for implementing the program in piano teaching sessions was developed, grounded in theoretical principles of creativity development according to flow theory, divergent and convergent thinking. The study involved 566 students from a music conservatory. Creativity was assessed using two forms of the Torrance Tests of Creative Thinking (TTCT). At the beginning of the pre-test, both groups demonstrated uniform levels of creativity within the music education process. Analysis of creativity revealed a significant increase in creative skills within the AI-intervention group ($p=0.000$). Considering the aim to more precisely determine the impact of recommendation algorithms on queries based on theoretical concepts of creativity development, a correlation was established between the frequency of application of the proposed query approaches and the dynamics of creativity between the pre- and post-tests ($r=0.001$). The results indicate notable improvements in creativity within the experimental group; the difference between the control and experimental groups increased to 23.1849, with the experimental group showing a mean of 67.311. Key concerns among students relate to the precise tuning of question-answer algorithms and the training required for effective use of technology for academic and creative development.

Keywords ChatGPT · Convergent thinking · Creativity · Divergent thinking · Flow theory · Musical education · Piano

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1 Introduction

Chinese musical art has undergone significant transformations over the centuries, at the core of the performance is piano playing (Guan, 2023). The perception of piano playing depends not only on technical skills but also on the repertoire, the choice of repertoire should be guided by the thematic focus of the concert, the pianist's technical skills, and the ultimate goal (whether to affect the audience's emotions or to win a competition) (Liu & Zhou, 2021). When selecting repertoire, the musician's professional capabilities should be considered, including education, imagination, artistic thinking, genre preferences, and the opportunity for creative expression (Hansen et al., 2016). To engage the audience, it is advisable to start with a simpler repertoire and progress to more complex compositions (Wong et al., 2021). Regardless of the complexity, the repertoire should include pieces with expressive musical elements, including artistic qualities (melodicity) and technical aspects (texture, tempo, and rhythm) (Bishop & Goebl, 2018). Several piano pieces within a single concert should be connected by a unified idea, emotions, storyline, or musical line to captivate the listeners. Arrangements can enhance the emotional impact on the audience (Mei & Yang, 2021). In performance, music should occupy the central role rather than serve as a background for depicting life situations through sound effects; classical music reveals the audience's emotionality and should aim at transforming thinking and perception of the surrounding world (Li et al., 2021). Regardless of the piano music genre, the pianist must convey the main content of the composition, along with its inherent emotionality and expressiveness (Zheng & Leung, 2021). All of this resonates with the effectiveness of the educational process, penetrated by a series of transformations and risks.

In the musical education of Chinese pianists, a crucial aspect is the balance between technical mastery and theoretical knowledge; reverence in this context encompasses not only technical skills, articulation, and related elements (Zheng & Leung, 2023). It is important to highlight the development of students' creative abilities as a foundation for their future musical careers (Bai, 2021). Like any artistic discipline, music demands significant creative effort from the musician, beyond merely creating their compositions, students must be capable of multitasking, seeking alternatives, standing out, and, most importantly, continually surprising their audience (Rui et al., 2023). Among familiar musical processes that can enhance creativity, contexts related to practice are often noted (Ho, 2023). For example, improvisation helps students develop the ability to make musical decisions on the fly. Interpretative approaches allow for bringing new perspectives to established genres or compositions (Lu, 2022).

A significant gap in the development of creativity among Chinese students remains the disconnect between practice and theory, with the latter focusing predominantly on the standardization of musicians (Zhang & Negus, 2020). The concurrent development of this skill contributes to a deeper understanding of the material, which directly impacts the musician's success, rather than merely memorizing composers, notes, and related elements, students learn to address musical problems, which stimulates their critical thinking (Wan, 2024). Creative approaches to theoretical exposition can enhance student motivation and engagement, and develop self-regulation skills even

during the reflection phase with the material (Yücetoker et al., 2021). One of the key arguments remains the complexity of understanding music; at the core of a creative approach is the truth of the interaction between theory and practice, where individual perception will shape the future musician's profile (Stefanova et al., 2021).

The advancing era of technology drives the community's focus on innovations as a key to addressing various challenges (Abdallah & Alkaabi, 2023; Chen, 2021; Cui, 2023; Ma & Ma, 2023; Song, 2024). However, there is a lack of standardized and varied approaches that dictate the principles for implementing AI as a genuinely effective tool for creativity development. This article aims to address this gap by providing context.

1.1 Problem statement

This study expands the research domain by analyzing the impact of generative artificial intelligence on music education. The relevance of this issue is underscored by the proactive integration of AI into various fields, including education. However, creative artistic education may face specific challenges in this regard. The analysis reveals that it is crucial to precisely delineate the aspects of AI's impact on the educational community and various criteria for academic achievements and skills (Abulibdeh et al., 2024; Yu, 2024). Recent research indicates that there is insufficient evidence to unanimously establish the positive impact of AI on music education (Li & Wang, 2024; Yu et al., 2023). Moreover, there is substantial debate regarding its influence on students' creative aspects, and scholars are discussing whether AI algorithms can enhance creative output (Marrone et al., 2022; Wang et al., 2023). Therefore, this article contributes arguments regarding the impact of technology on this significant predictor of future success in music.

Objective of the Study: To investigate the impact of ChatGPT on music education by evaluating creativity and perception indicators resulting from AI integration.

Research Objectives:

- Measure the level of creativity among music students.
- Integrate interactions with the artificial intelligence ChatGPT into the educational curriculum.
- Assess students' perceptions of the implemented program.
- Develop an understanding of the dynamics of the studied variable following AI integration.

1.2 Literature review

The professionalism of playing piano compositions lies in technique improvement and study of various musical pieces with transitions to a more complex piano repertoire. This approach contributes to the empowerment of musicians and the audience's interest in listening to compositions. The expressive means of composition influence the acquisition of the necessary emotions and experiences by the audience. Emotional intelligence influences overall socialization (Zhukov & Rowley, 2022). The study of the piano repertoire made it possible to identify characteristic elements that have an

impact on listeners using descriptive analysis. It has been established that the audience first perceives musical elements and evaluates musical expressions, tempo, and rhythm. Emphasis on these indicators while playing the piano contributes to the audience's interest (Gurler, 2022). An analysis of Howard Skempton's piano music has revealed that it is characterized by simplicity and sophistication as a result of repetitive patterns. Listening to his music relieves tension, harmonizes resonant absorption and positive mood, and increases creative and labor activity (Cavett, 2022).

Also, the quality of music perception depends on the manner of performance, which is characterized by personal characteristics such as timbre, intonation, emotionality, sound accents, and pauses (Bland & Cho, 2021). The movements of pianists also influence the performance manner and the expressiveness of the musical interpretation. Chinese piano concertos are based on the transmission of general themes and not on general forms of sound, which are interconnected with the means of expressiveness and melody (Wang, 2018). Also, during performances, metro-rhythmic themes are taken into account; they are interconnected with intonation and rhythmic formulas expressed in Chinese musical turns (Yang et al., 2023). The performance manner also depends on the timbre-textural features that affect the change in melodies, the sound color, and brightness (Yang, 2021). The uniqueness of creating melodies is associated with a unique sound intonation and rhythm, which are interconnected with melodic themes (Harrison, 2021). The European approaches to piano performance in Chinese music contribute to the manifestation of new artistic solutions, which affect the audience's interest and changes in intonation features (Bland & Cho, 2021).

The involvement of conductors in musical concerts contributes to the greater expressiveness of the music sounds. The brightness and immediacy of compositions help to attract more listeners. This is based on the creation of psycho-emotional relief and interaction with other people. Conductors have an influence on the formation of different emotions while listening to the same musical composition as a result of a different approach to performance: at a fast or slow tempo, in a changed key, etc. (Poggi et al., 2021). The individual style of playing piano compositions is based on the variation of tempo (base, global, and local). The results show that a change in the tempo of musical compositions affects the change in the audience's emotions and forms various experiences. Listening to music also affects the development of creative skills and increases independent activities (Zhou & Fabian, 2021). Musical competitions are aimed at the development of talented contestants by staging performances, using a diverse repertoire, and controlling emotions. A diverse repertoire impacts not only the audience but also the jury because repertoire can reveal professional skills and provide emotionality and colorful performances. Music competitions contribute to the development of personality and influence the disclosure of creative and technical skills (van Wyk & de Villiers, 2021).

The musical repertoire should be formed under the principles of excellence and artistic expression. The repertoire choice depends on the audience; some prefer modern compositions, and others prefer traditional ones. An important feature of the repertoire choice is its progressiveness, which will influence the development of musicians' technical skills and help attract a larger audience. Repertoire choice should take into account its artistic and technical potential. Complex performance

techniques affect the melody and emotionality of the composition (Sternberg & Kibelsbeck, 2022). The metrical structure of a musical composition affects the change in its emotionality. Aesthetic perception of the composition depends on its artistic value and performance virtuosity. During the performance, improvisational elements should be used that affect genre expressiveness. Scales, notes, pitch, and tonal modulation affect musical expressiveness. The music tonality depends on the scales and the level of their attraction to the original note (Breen & Fitzroy, 2021). Melodic rhythm affects the drama of musical art in musical theater. The musical repertoire influences melody and lyricism. The fusion of music and theater contributes to creating a special atmosphere and enlivening concerts that attract listeners. Various musical modulations and sound combinations evoke multiple emotions. The music harmony affects tension and brightness. Music changes affect the complexity of emotional experiences and the melody perception by the audience (Broughton et al., 2021).

Scientific studies have already demonstrated successful cases of AI implementation in enhancing music education. For instance, the integration of AI has led to a 15% improvement in academic performance; however, variations in learning intensity have resulted in disparities in achievement. Additionally, piano performance improved by 6.51%, while academic scores in solfeggio and music literature increased by 4% (Li & Wang, 2024). Furthermore, through the integration of a multi-user AI detection algorithm, an effective model for online music education design was established. An analysis of the findings indicated that an AI-based online music education system can significantly enhance the efficiency of music learning for the target audience and offers evident advantages for student groups (Yu et al., 2023).

In another study, the authors concluded that AI can create a more optimized environment and professional music classes, enabling instructors and students to maximize its potential and ensure a seamless improvement of the online learning model. The study reported that student learning outcomes reached 95.2%, while the efficiency coefficient was 98.1% (Wei et al., 2022).

2 Methods

2.1 Research design

The study was based on analyzing the impact of AI integrated into music education. To this end, an approach was developed for implementing the program in piano teaching sessions. It is important to note that the educational system at the examined university involved a division between theoretical and practical classes. The developed program involved full integration into the lecture component of piano instruction for the experimental group of respondents.

The program lasted for one academic year (2023–2024). Unlike the traditional approach (used in the control group), the introduced program involved balanced interaction with theoretical material from both students and teachers, as well as AI technology. During half of the lecture time, students listened to the assigned topic within the framework of the university's methodological study. In the other half, students interacted with ChatGPT. They could refine their notes on the subject according

to their queries, clarify examples of specific compositions or techniques, and so on. Additionally, students worked on a topic different from previous sessions, aiming to deepen their understanding within this timeframe.

The history of queries was stored on the training computer. Students had the opportunity to interact with the technology based on their own experience or to use prompts. Teachers provided students with instructions on using this technology and a list of effective algorithms or guidelines for ChatGPT. The recommendations were based on more recent research (Božić & Poola, 2023; Sharma & Yadav, 2022), professional guidance from AI developers, consultations with computer science doctors, and so forth. The recommendations were formulated based on prominent theories of creativity development. For instance, the authors drew on Mihály Csíkszentmihályi's Flow Theory, which, in the context of individualized queries, helps bring individuals closer to a state of flow by applying personalized prompts and generating audio and video supplementary materials (Flores et al., 2019). Home assignments included aspects that required student improvisation and skills gained from lecture sessions. The recommendations also incorporated the contexts of divergent and convergent thinking, which, in response to algorithms from recommended materials, teach students to sort task complexities and apply various approaches, thereby enhancing their analytical and creative thinking skills (Koivisto & Grassini, 2023).

Table 1 presents examples of algorithms used during consultations with students. Thus, the proposed queries for students could utilize AI parameters to foster creativity through various approaches. To ensure that the control group did not use artificial intelligence outside the experimental environment, a comprehensive approach was implemented, integrating administrative, educational, and motivational strategies. Participants in the control group received explicit instructions prohibiting the use of any AI tools while completing tasks related to the study. To enhance awareness of the importance of compliance with these guidelines, explanatory discussions were conducted, emphasizing their critical role in ensuring the scientific validity of the experiment's results.

To monitor adherence to the established requirements, regular surveys and self-reports were employed, allowing for the identification of potential violations. Additionally, the instructional process was structured in such a way that the majority of tasks were completed directly in the classroom under the supervision of instructors, thereby minimizing the risk of uncontrolled AI use.

The motivational component also played a crucial role: participants were informed about the significance of their contribution to the research project, fostering a sense of responsibility and commitment to the experiment.

2.2 Survey

To measure creativity before and after the intervention, two forms of the Torrance Tests of Creative Thinking (TTCT) were used to assess human creative thinking. This test includes Form A and Form B, and the scoring system comprises five subscales: fluency, originality, elaboration, abstractness of titles, and resistance to premature closure (Clapham, 1998). The test has the following scoring norms: 30 points– poor; 30–34 points– below average; 35–39 points– slightly below average; 40–60 points–

Table 1 Examples of recommendations for enhancing creativity through ChatGPT in music education

No	Recommendation	Theoretical Basis	Example Query
1	Use different levels of detail in responses to gain a deeper understanding of the topic	Convergent Thinking: From simple to complex concepts	*Compare different types of chords: Provide examples from nature, art*, etc.
2	Mix queries between open and closed. Formulate open questions to stimulate discussion	Divergent Thinking: Generating ideas based on different perspectives, many ideas	*I disagree with the statement that the Suetang school and the "Yuege" educational system laid the foundations for contemporary Chinese musical culture. In my opinion, these positions are now outdated and do not meet the needs of audiences and performers. Present your own arguments for and against such a statement.*
3	Ask about the practical application of theory	Flow Theory: Personal experience, the intersection between theoretical and practical aspects	*How can I use the concept of counterpoint in piano compositions?*

average; 61–65 points– slightly above average; 66–70 points– above average; and over 70 points indicates very high creative thinking ability. The test applies to adults and students (Coronado Hijón, 2015; Lokki et al., 2016).

The study also assessed students' perceptions of a specific technology. For this purpose, the authors employed the approach outlined in Shoufan (2023). The questionnaire consists of 27 questions with responses on a Likert scale (from 1 to 5). It is based on 15 themes defining technology perception contexts, divided into positive and negative blocks. Responses to negative questions were reverse-coded on the Likert scale.

2.3 Sample

The sample consisted of 566 students from the Shanghai Conservatory of Music. The students were majoring in piano in their second and third years. The average age of the students was 23.9 (SD = 7.88; SEM = 1.838); by gender, the distribution included 394 female students and 173 male students. The students were divided into two equal groups of 283 using a randomizer to ensure the experiment's integrity.

2.4 Statistical processing

The analysis was carried out using IBM SPSS 20 software.

2.5 Ethical issues

Ethical considerations are related to adherence to publication rules to prevent the use of borrowed information. Students consented to the processing of their personal data. They attended a lecture on the use of artificial intelligence and were informed of the experimental guidelines. A supervisor was present at each session to oversee the progress of the experiment.

2.6 Research limitations

The program focused exclusively on the music specialty and was based on fostering creativity. The sample was drawn from China. Additionally, it is important not to generalize the results according to specific theories underpinning the intervention. The experiment was specifically based on ChatGPT technology. External factors that could influence creativity, such as family, hobbies, etc., were not considered. Students might have had varying experiences with AI, which could also have impacted the experiment's results.

3 Results

As shown in Table 2, the difference between the control and experimental groups can be considered insignificant. This is indicated by the statistical significance results from the t-test.

The result between both groups is characterized by an equality of means of 9.2367, and with a p-value greater than 0.05, it demonstrates the absence of significant differences. This indicates that the pre-test values for both groups are aligned in the context of creativity level analysis. It is worth noting that both results, in the context of the control and experimental groups, fall within the average normative values, yet neither group exhibited a significant level of creativity.

Table 2 Analysis of statistical significance of creativity differences between control and experimental groups before the intervention

Level of creativity								95% Confidence Interval	
	F	Sig.	t	DF	Sig. (2 tailed)	Difference of Means	Standard Error of the Difference	Lower bound	Upper bound
Equal variances assumed	150.3674	0.6299	0.2748	564	0.1941	9.2367	0.092	8.003	10.728
Equal variances not assumed			0.2748	560.399	0.1941	9.2367	0.092	8.874	10.849

Table 3 Analysis of statistical significance of creativity differences between control and experimental groups after the intervention

Level of creativity								95% Confidence Interval	
	F	Sig.	t	Df	Sig. (2-tailed)	Difference of Means	Standard Error of the Difference	Lower bound	Upper bound
Equal variances assumed	628.209	0.000	10.688	564	0.000	23.1849	0.077	23.001	23.748
Equal variances not assumed			10.688	560.399	0.000	23.1849	0.077	22.993	23.162

Table 4 Correlation between provided recommendations and changes in creativity levels

		Level of Creativity (Difference Before and After the Intervention)
Frequency of Recommendation Usage in the Course	r	0.001
	Sig. (2 tailed)	0.000
	N	566

For the analysis of dynamics, attention should be directed to the post-intervention indicators of the artificial intelligence program, as presented in Table 3.

The results indicate notable improvements in creativity within the experimental group. The difference between the control and experimental groups increased to 23.1849, with the experimental group showing a mean of 67.311, indicating significantly high values above the norm. The significance level of the results was found to be within the acceptable range ($p=0.000$), suggesting that the AI program had an impact on musical creativity.

It is important to note that, when analyzing aspects and following recommendations, the correlation with applying all theoretical recommendations in the work should be considered (see Table 4).

In this context, it is noteworthy that the frequency of utilizing the proposed approach at least once per session correlates with the highest creativity scores among students in the experimental group. Thus, it can be asserted that the developed approach effectively influences the creativity of piano students, as an increased frequency of recommendation usage corresponds with greater improvements in creativity.

On the other hand, a more detailed examination of the students' perceptions of such innovations is warranted (see Table 5).

Analysis of the table suggests that the predominant opinion among respondents focuses on the positive perception of the technology within the context of its utility in the musical educational process. The positive aspects should be attributed to the assessment of AI as a valuable and effective educational resource, with usefulness for work rated at 4.77 and for learning at 4.71. Students also unanimously highlighted the quality of explanation (4.89), indicating the effectiveness of generative algorithms

Table 5 Analysis of students' perceptions of ChatGPT

	Mean	SD	SEM
PT1: Enthusiasm and Gratitude	4.3	0.0368	0.00013
PT2: Useful for Learning	4.71	0.0628	0.00022
PT3: Human-Like Language	4.11	0.0227	0.00008
PT4: Useful for Work	4.77	0.0343	0.00012
PT5: Interesting and Motivating	4.22	0.0638	0.00022
PT6: Good Explanation	4.89	0.0368	0.00013
PT7: Optimism	4.37	0.0567	0.0002
PT8: Easy to Use	4.45	0.0512	0.0001
	4.55	0.0937	0.00033
NT1: Inaccurate Responses	3.51	0.0573	0.0002
NT2: Cannot Replace Human Intelligence	4.51	0.0753	0.00026
NT3: Negative Impact on Education	4.01	0.0916	0.00032
NT4: Difficult to Use	4.16	0.0868	0.0003
NT5: Potential for Malicious Use	3.93	0.0738	0.00026
NT6: May Threaten Job Opportunities	4.29	0.0582	0.0002

in addressing gaps in theory and providing personalized responses to queries. The greatest risks, according to the perception analysis, are related to concerns about the complete replacement of human intelligence (3.51) and the complexity of use (4.01). This is particularly relevant to formulating queries and understanding AI's responses.

Given this analysis, it is important to note that the influence of AI based on ChatGPT can be considered positive. The perception of this intervention is also characterized by predominant positive perspectives from the students.

4 Discussion

Considering the applied approach of teacher prompts based on logical algorithms of queries in accordance with theories of creativity development, it is pertinent to compare the obtained results with existing approaches in the literature. Highlighting the positive impact of AI, it is important to mention the potential use of the flipped classroom model and its influence on creativity. Lv (2023) emphasizes the key reactions to the impact of the presence of clear and timely feedback on their musical works, which helped them better understand their mistakes and find new ways to express their ideas. This influence encouraged students to maximize the effect of the queries received and to develop creative solutions to improve their work. The flipped classroom model using AI allowed instructors to devote more time to individual work with students and assist in developing their creative skills. This article also highlights individualized approaches as fundamental to fostering creativity in music education. Among the primary challenges in implementing such approaches within a traditional system is the issue of working with general lecture groups, where, for various reasons, personalized approaches may not be provided by the educator. According to the principles of prompts based on flow theory, this component was also emphasized in the course. Notably, student interaction with an intelligent assistant allows them to expand their horizons and enhance their creative capabilities. Zhang et al. (2024)

found that AI composition software provided students with new opportunities for experimentation and facilitated a deeper understanding of musical principles. In this context, the significance of this tool for the theoretical foundations of music study is highlighted as a valuable source of new information, synthesis of approaches, and generation of new ideas.

Additionally, the important contribution of AI to students' self-regulation processes should be noted. Contemporary conditions necessitate acquiring knowledge independently of full reliance on educators, where AI can serve as an auxiliary source of knowledge (Delima et al., 2024). Musicians can focus on their creativity, with technology serving as a supplementary tool for advice and enhancing creative potential (Wei et al., 2022).

The scope of impact can be expanded through practical applications. Students may explore new genres in musical performance that are not covered in standard practice. Thus, AI facilitates reflection on their work (Dai, 2021). It is also important to note that a potentially additional positive effect of using AI is the contemporary dependence of youth on innovative technologies, which further encourages technology integration into the music education process. The approaches described above are vividly illustrated in similar cases by Zulić (2019), which further emphasizes the relevance of using AI in music education.

Analyzing the key components of ChatGPT, several compelling arguments can be made regarding its effectiveness in fostering creativity. As one of the most widely used generative AIs globally, it plays a significant role in resolving functional issues and managing large datasets daily. This underscores the importance of its continuous data updating and adjustment to work with parameters familiar to students (Holster, 2024). In the context of student perception, its adaptability is crucial and should be continuously improved. The functional interface of this technology enables easy and accessible interaction, which enhances the focus on student creativity rather than complicating the task (Zhou & Kim, 2024). A major challenge and gap in using ChatGPT is ensuring the relevance of responses and the accuracy of queries. Therefore, this study focused on recommendations rather than, for example, the integration format of AI, which is the focus of many other studies. Working with theoretical approaches can help mitigate the potential negative effects of chaotic and disorganized queries. While structuring opinions as a supporting premise for enhancing creativity is addressed in several articles, it is crucial to balance the use of technology with creative abilities. Some argue that excessive reliance on AI could lead to standardization (Vidani, 2024). Hence, it is important to consider influencing factors more thoroughly, providing guidance for future researchers.

5 Conclusion

The study successfully addressed all the set objectives. Analysis of creativity revealed a significant enhancement in creative skills within the intervention group using the author's approach to AI ($p = 0.000$). In contrast, the control group showed no notable changes, while both groups exhibited uniform levels of creativity at the outset of the pre-test within the context of music education.

Given the aim to more precisely assess the impact of recommendation algorithms based on theoretical principles of creativity development, the correlation indicator demonstrates a significant relationship between the frequency of applying the proposed approaches to queries and the dynamics of creativity between pre- and post-tests ($r=0.001$).

The students' perceptions of AI indicate the potential for more comprehensive and systematic implementation. Major concerns among students highlight the need for more precise algorithms for question-answer interactions and the necessity of educating students on the correct utilization of technological tools for academic and creative advancement.

The theoretical significance of this article lies in expanding the scope of AI's influence on the field of music education. The experiments utilized a program based on theories of creativity development to enhance the integration of innovations with existing principles of psychology and pedagogy. This study contributes to bridging the gap in understanding the impact of ChatGPT on music education in China and identifying potential weaknesses in such collaborations.

The practical significance of the article is found in the development of a model for interaction between students and AI, which has demonstrably impacted effectiveness. Potential barriers to the effective implementation of ChatGPT were identified, and key points for enhancing the integration of technology into education were established. The article is relevant to students, music educators, policymakers, and AI developers.

Future researchers may focus on exploring other variables influenced by the program. Investigating prompt engineering (PE) as a means to enhance the accuracy of formulations and refine student responses could prove valuable. Moreover, examining the impact of external factors on creativity development and their correlation within the context of music education warrants further study.

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Declarations

Ethics approval and consent to participate The research was conducted ethically in accordance with the World Medical Association Declaration of Helsinki. The research was approved by the local ethics committees of Taiyuan Normal University.

Informed consent Informed consent was obtained from all subjects involved in the study.

Consent for publication Not applicable.

Conflicts of interest This research has no conflict of interests.

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